THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The atomic mass of oxygen is: 1

(a) 16 u **(b)** 12 u **(c)** 32 u **(d)** 8 u

2. A molecule of water has: 1

(a) 4 atoms **(b)** 2 atoms **(c)** 1 atom **(d)** 3 atoms

3. The formula of calcium oxide is: 1

(a) Ca_2O_2 **(b)** CaO_2 **(c)** Ca_2O **(d)** CaO

4. The molecular mass of O_2 is: 1

(a) 64 u **(b)** 32 u **(c)** 16 u **(d)** 8 u

5. The formula of carbon dioxide is: 1

(a) C_2O_2 **(b)** C_2O **(c)** CO **(d)** CO_2

6. The formula of hydrogen chloride is: 1

(a) HCl_2 **(b)** H_2Cl_2 **(c)** H_2Cl **(d)** HCl

7. The molecular mass of H_2 is: 1

(a) 4 u **(b)** 16 u **(c)** 2 u **(d)** 1 u

8. The formula of ammonia is: 1

(a) N_2H **(b)** NH_4 **(c)** N_3H **(d)** NH_3

9. The formula of hydrogen sulphide is: 1

(a) HS **(b)** HS_2 **(c)** H_2S_2 **(d)** H_2S

10. The molecular mass of CO_2 is: 1

(a) 44 u **(b)** 28 u **(c)** 22 u **(d)** 32 u

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the molecular mass of H_2O. 2

12. Define atomicity. Give the atomicity of O_2. 2

13. Name the bond formed by sharing of electrons. 2

14. Calculate the formula unit mass of $NaCl$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the formulae: sodium oxide, magnesium chloride, calcium oxide. 3

16. Give three examples of molecules of compounds. 3

17. State the mass ratio of hydrogen to oxygen in water and the mass of oxygen in 18 g of water. 3

OR

List three properties of ionic compounds. 3

18. How do atoms of a metal form ions? Give two examples. 3

OR

Write the formula of three ionic compounds. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the formation of the hydrogen molecule. 5

OR

Explain the properties of ionic compounds. 5

20. Differentiate between ionic and covalent bonds. 5

OR

Explain the properties of covalent compounds. 5

END OF QUESTION PAPER — Total: 40 Marks

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